

Troubleshooting

Chapter 4 provides information to guide you in troubleshooting the PowerFlex 4 drive. Included is a listing and description of drive faults (with possible solutions, when applicable).

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Drive Status

The condition or state of your drive is constantly monitored. Any changes will be indicated through the integral keypad.

LED Indications

See [page 2-3](#) for information on drive status indicators and controls.

Faults

A fault is a condition that stops the drive. There are two fault types.

Type	Fault Description	
①	Auto-Reset/Run	When this type of fault occurs, and A092 [Auto Rstrt Tries] is set to a value greater than "0," a user-configurable timer, A093 [Auto Rstrt Delay], begins. When the timer reaches zero, the drive attempts to automatically reset the fault. If the condition that caused the fault is no longer present, the fault will be reset and the drive will be restarted.
②	Non-Resetable	This type of fault may require drive or motor repair, or is caused by wiring or programing errors. The cause of the fault must be corrected before the fault can be cleared.

Fault Indication

Condition	Display
Drive is indicating a fault. The integral keypad provides visual notification of a fault condition by displaying the following. - Flashing fault number - Flashing fault indicator Press the Escape key to regain control of the integral keypad.	

Manually Clearing Faults

Step	Key(s)
- Press Esc to acknowledge the fault. The fault information will be removed so that you can use the integral keypad. Access d007 [Fault 1 Code] to view the most recent fault information. - Address the condition that caused the fault. The cause must be corrected before the fault can be cleared. See Table 4.A. - After corrective action has been taken, clear the fault by one of these methods. - Press Stop if P037 [Stop Mode] is set to a value between “0” and “3”. - Cycle drive power. - Set A100 [Fault Clear] to “1” or “2”. - Cycle digital input if A051 - A052 [Digital Inx Select] is set to option 7 “Clear Fault”.	

Automatically Clearing Faults

Option / Step
Clear a Type 1 fault and restart the drive. - Set A092 [Auto Rstrt Tries] to a value other than “0”. - Set A093 [Auto Rstrt Delay] to a value other than “0”.
Clear an OverVoltage, UnderVoltage or Heatsink OvrTmp fault without restarting the drive. - Set A092 [Auto Rstrt Tries] to a value other than “0”. - Set A093 [Auto Rstrt Delay] to “0”.

Auto Restart (Reset/Run)

The Auto Restart feature provides the ability for the drive to automatically perform a fault reset followed by a start attempt without user or application intervention. This allows remote or “unattended” operation. Only certain faults are allowed to be reset. Certain faults (Type 2) that indicate possible drive component malfunction are not resettable.

Caution should be used when enabling this feature, since the drive will attempt to issue its own start command based on user selected programming.

Fault Descriptions

Table 4.A Fault Types, Descriptions and Actions

No.	Fault	Type ⁽¹⁾	Description	Action
F2	Auxiliary Input	①	Auxiliary input interlock is open.	1. Check remote wiring. 2. Verify communications programming for intentional fault.
F3	Power Loss	②	DC bus voltage remained below 85% of nominal.	1. Monitor the incoming AC line for low voltage or line power interruption. 2. Check input fuses.
F4	UnderVoltage	①	DC bus voltage fell below the minimum value.	Monitor the incoming AC line for low voltage or line power interruption.
F5	OverVoltage	①	DC bus voltage exceeded maximum value.	Monitor the AC line for high line voltage or transient conditions. Bus overvoltage can also be caused by motor regeneration. Extend the decel time or install dynamic brake option.
F6	Motor Stalled	①	Drive is unable to accelerate motor.	Increase P039 - A067 [Accel Time x] or reduce load so drive output current does not exceed the current set by parameter A089 [Current Limit].
F7	Motor Overload	①	Internal electronic overload trip.	1. An excessive motor load exists. Reduce load so drive output current does not exceed the current set by parameter P033 [Motor OL Current]. 2. Verify A084 [Boost Select] setting
F8	Heatsink OvrTmp	①	Heatsink temperature exceeds a predefined value.	1. Check for blocked or dirty heat sink fins. Verify that ambient temperature has not exceeded 40°C (104°F) for IP 30/NEMA 1/UL Type 1 installations or 50°C (122°F) for IP20/Open type installations. 2. Check fan.
F12	HW OverCurrent	②	The drive output current has exceeded the hardware current limit.	Check programming. Check for excess load, improper A084 [Boost Select] setting, DC brake volts set too high or other causes of excess current.
F13	Ground Fault	②	A current path to earth ground has been detected at one or more of the drive output terminals.	Check the motor and external wiring to the drive output terminals for a grounded condition.

⁽¹⁾ See [page 4-1](#) for a description of fault types.

No.	Fault	Type ⁽¹⁾	Description	Action
F33	Auto Rstrt Tries	②	Drive unsuccessfully attempted to reset a fault and resume running for the programmed number of A092 [Auto Rstrt Tries].	Correct the cause of the fault and manually clear.
F38	Phase U to Gnd	②	A phase to ground fault has been detected between the drive and motor in this phase.	1. Check the wiring between the drive and motor. 2. Check motor for grounded phase. 3. Replace drive if fault cannot be cleared.
F39	Phase V to Gnd			
F40	Phase W to Gnd			
F41	Phase UV Short	②	Excessive current has been detected between these two output terminals.	1. Check the motor and drive output terminal wiring for a shorted condition. 2. Replace drive if fault cannot be cleared.
F42	Phase UW Short			
F43	Phase VW Short			
F48	Params Defaulted		The drive was commanded to write default values to EEPROM.	1. Clear the fault or cycle power to the drive. 2. Program the drive parameters as needed.
F63	SW OverCurrent	①	Programmed A098 [SW Current Trip] has been exceeded.	Check load requirements and A098 [SW Current Trip] setting.
F64	Drive Overload	②	Drive rating of 150% for 1 minute or 200% for 3 seconds has been exceeded.	Reduce load or extend Accel Time.
F70	Power Unit	②	Failure has been detected in the drive power section.	1. Cycle power. 2. Replace drive if fault cannot be cleared.
F81	Comm Loss	②	RS485 (DSI) port stopped communicating.	1. If adapter was not intentionally disconnected, check wiring to the port. Replace wiring, port expander, adapters or complete drive as required. 2. Check connection. 3. An adapter was intentionally disconnected. 4. Turn off using A105 [Comm Loss Action].
F100	Parameter Checksum	②	The checksum read from the board does not match the checksum calculated.	Set P041 [Reset To Defaults] to option 1 "Reset Defaults".
F122	I/O Board Fail	②	Failure has been detected in the drive control and I/O section.	1. Cycle power. 2. Replace drive if fault cannot be cleared.

⁽¹⁾ See [page 4-1](#) for a description of fault types.

Common Symptoms and Corrective Actions

Motor does not Start.

Cause(s)	Indication	Corrective Action
No output voltage to the motor.	None	Check the power circuit. • Check the supply voltage. • Check all fuses and disconnects. Check the motor. • Verify that the motor is connected properly. Check the control input signals. • Verify that a Start signal is present. If 2-Wire control is used, verify that either the Run Forward or Run Reverse signal is active, but not both. • Verify that I/O Terminal 01 is active. • Verify that P036 [Start Source] matches your configuration. • Verify that A095 [Reverse Disable] is not prohibiting movement.

Drive does not Start from Start or Run Inputs wired to the terminal block.

Cause(s)	Indication	Corrective Action
Drive is Faulted	Flashing red status light	Clear fault. • Press Stop • Cycle power • Set A100 [Fault Clear] to option 1 "Clear Faults". • Cycle digital input if A051 - A052 [Digital Inx Select] is set to option 7 "Clear Fault".
Incorrect programming. • P036 [Start Source] is set to option 0 "Keypad" or option 5 "RS485 (DSI) Port". • A051 - A052 [Digital Inx Select] is set to option 5 "Local" and the input is active.	None	Check parameter settings.
Incorrect input wiring. See 1-14 for wiring examples. • 2 wire control requires Run Forward, Run Reverse or Jog input. • 3 wire control requires Start and Stop inputs • Stop input is always required.	None	Wire inputs correctly and/or install jumper.
Incorrect Sink/Source DIP switch setting.	None	Set switch to match wiring scheme.

Drive does not Start from Integral Keypad.

Cause(s)	Indication	Corrective Action
Integral keypad is not enabled.	Green LED above Start key is not illuminated.	- Set parameter P036 [Start Source] to option 0 "Keypad". - Set parameter A051 - A052 [Digital Inx Select] to option 5 "Local" and activate the input.
I/O Terminal 01 "Stop" input is not present.	None	Wire inputs correctly and/or install jumper.

Drive does not respond to changes in speed command.

Cause(s)	Indication	Corrective Action
No value is coming from the source of the command.	The drive "Run" indicator is lit and output is 0 Hz.	- Check d012 d012 [Control Source] for correct source. - If the source is an analog input, check wiring and use a meter to check for presence of signal. - Check d002 [Commanded Freq] to verify correct command.
Incorrect reference source is being selected via remote device or digital inputs.	None	- Check d012 [Control Source] for correct source. - Check d014 [Dig In Status] to see if inputs are selecting an alternate source. Verify settings for A051 - A052 [Digital Inx Select]. - Check P038 [Speed Reference] for the source of the speed reference. Reprogram as necessary. - Review the Speed Reference Control chart on page 1-17.

Motor and/or drive will not accelerate to commanded speed.

Cause(s)	Indication	Corrective Action
Acceleration time is excessive.	None	Reprogram P039 [Accel Time 1] or A067 [Accel Time 2].
Excess load or short acceleration times force the drive into current limit, slowing or stopping acceleration.	None	Compare d003 [Output Current] with A089 [Current Limit]. Remove excess load or reprogram P039 [Accel Time 1] or A067 [Accel Time 2]. Check for improper A084 [Boost Select] setting.
Speed command source or value is not as expected.	None	Verify d002 [Commanded Freq]. Check d012 [Control Source] for the proper Speed Command.
Programming is preventing the drive output from exceeding limiting values.	None	Check P035 [Maximum Freq] to insure that speed is not limited by programming.

Motor operation is unstable.

Cause(s)	Indication	Corrective Action
Motor data was incorrectly entered.	None	1. Correctly enter motor nameplate data into P031, P032 and P033. 2. Enable A097 [Compensation]. 3. Use A084 [Boost Select] to reduce boost level.

Drive will not reverse motor direction.

Cause(s)	Indication	Corrective Action
Digital input is not selected for reversing control.	None	Check [Digital Inx Sel] (See page 3-13). Choose correct input and program for reversing mode.
Digital input is incorrectly wired.	None	Check input wiring. (See page 1-13)
Motor wiring is improperly phased for reverse.	None	Switch two motor leads.
Reverse is disabled.	None	Check A095 [Reverse Disable].

Notes:

Supplemental Drive Information

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Drive, Fuse & Circuit Breaker Ratings

The tables on the following pages provide drive ratings and recommended AC line input fuse and circuit breaker information. Both types of short circuit protection are acceptable for UL and IEC requirements. Sizes listed are the recommended sizes based on 40 degree C and the U.S. N.E.C. Other country, state or local codes may require different ratings.

Fusing

If fuses are chosen as the desired protection method, refer to the recommended types listed below. If available amp ratings do not match the tables provided, the closest fuse rating that exceeds the drive rating should be chosen.

- IEC – BS88 (British Standard) Parts 1 & 2⁽¹⁾, EN60269-1, Parts 1 & 2, type gG or equivalent should be used.
- UL – UL Class CC, T or J must be used.⁽²⁾

Circuit Breakers

Refer to listings in the following tables for recommended circuit breakers (inverse time or instantaneous trip) and 140M Self-Protecting Motor Starters.

⁽¹⁾ Typical designations include, but may not be limited to the following; Parts 1 & 2: AC, AD, BC, BD, CD, DD, ED, EFS, EF, FF, FG, GF, GG, GH.

⁽²⁾ Typical designations include; Type CC - KTK-R, FNQ-R
Type J - JKS, LPJ
Type T - JJS, JJN

Specifications

Drive Ratings

Catalog Number	Output Ratings		Input Ratings			Branch Circuit Protection			Power Dissipation IP20 Open Watts
	kW (HP)	Amps	Voltage Range	kVA	Amps	Fuses	140M Motor Protectors ⁽²⁾	Contactors	
100 - 120V AC (±10%) – 1-Phase Input, 0 - 230V 3-Phase Output									
22A-V1P5N104	0.2 (0.25)	1.5	90-126	0.75	6.0	10	140M-C2E-C10	100-C09	25
22A-V2P3N104	0.4 (0.5)	2.3	90-126	1.15	9.0	15	140M-C2E-C16	100-C12	30
22A-V4P5N104	0.75 (1.0)	4.5	90-126	2.25	18.0	30	140M-D8E-C20	100-C23	50
22A-V6P0N104	1.1 (1.5)	6.0	90-126	3.0	24.0	40	140M-D8E-C25	100-C37	70
200 - 240V AC (±10%) – 1-Phase ⁽¹⁾ Input, 0 - 230V 3-Phase Output, NO BRAKE									
22A-A1P4N103	0.2 (0.25)	1.4	180-265	0.7	3.2	6	140M-C2E-B40	100-C09	25
22A-A2P1N103	0.4 (0.5)	2.1	180-265	1.05	5.3	10	140M-C2E-B63	100-C09	30
22A-A3P6N103	0.75 (1.0)	3.6	180-265	1.8	9.2	15	140M-C2E-C16	100-C12	50
22A-A6P8N103	1.5 (2.0)	6.8	180-265	3.4	14.2	25	140M-C2E-C16	100-C16	75
22A-A9P6N103	2.2 (3.0)	9.6	180-265	4.8	19.6	30	140M-D8E-C25	100-C23	100
200 - 240V AC (±10%) – 1-Phase ⁽¹⁾ Input, 0 - 230V 3-Phase Output									
22A-A1P5N104	0.2 (0.25)	1.5	180-265	0.75	5.0	10	140M-C2E-B63	100-C09	25
22A-A2P3N104	0.4 (0.5)	2.3	180-265	1.15	6.0	10	140M-C2E-B63	100-C09	30
22A-A4P5N104	0.75 (1.0)	4.5	180-265	2.25	10.0	15	140M-C2E-C16	100-C12	50
22A-ABP0N104	1.5 (2.0)	8.0	180-265	4.0	18.0	30	140M-D8E-C20	100-C23	80
200 - 240V AC (±10%) – 3-Phase Input, 0 - 230V 3-Phase Output									
22A-B1P5N104	0.2 (0.25)	1.5	180-265	0.75	1.8	3	140M-C2E-B25	100-C09	25
22A-B2P3N104	0.4 (0.5)	2.3	180-265	1.15	2.5	6	140M-C2E-B40	100-C09	30
22A-B4P5N104	0.75 (1.0)	4.5	180-265	2.25	5.2	10	140M-C2E-C10	100-C09	50
22A-B8P0N104	1.5 (2.0)	8.0	180-265	4.0	9.5	15	140M-C2E-C16	100-C12	80
22A-B012N104	2.2 (3.0)	12.0	180-265	5.5	15.5	25	140M-C2E-C16	100-C16	115
22A-B017N104	3.7 (5.0)	17.5	180-265	8.6	21.0	35	140M-F8E-C25	100-C23	165
380 - 480V AC (±10%) – 3-Phase Input, 0 - 460V 3-Phase Output									
22A-D1P4N104	0.4 (0.5)	1.4	340-528	1.4	1.8	3	140M-C2E-B25	100-C09	30
22A-D2P3N104	0.75 (1.0)	2.3	340-528	2.3	3.2	6	140M-C2E-B40	100-C09	40
22A-D4P0N104	1.5 (2.0)	4.0	340-528	4.0	5.7	10	140M-C2E-B63	100-C09	60
22A-D6P0N104	2.2 (3.0)	6.0	340-528	5.9	7.5	15	140M-C2E-C10	100-C09	90
22A-D8P7N104	3.7 (5.0)	8.7	340-528	8.6	9.0	15	140M-C2E-C16	100-C16	145

Input/Output Ratings	Approvals
Output Frequency: 0-240 Hz (Programmable) Efficiency: 97.5% (Typical)	  EMC Directive 89/336/EEC, LV Dir. 73/23/EEC LV: EN 50178 EMC: EN 61800-3

Digital Control Inputs (Input Current = 6mA)	Analog Control Inputs
SRC (Source) Mode: 18-24V = ON 0-6V = OFF	SNK (Sink) Mode: 0-6V = ON 18-24V = OFF
4-20mA Analog: 250 ohm input impedance 0-10V DC Analog: 100k ohm input impedance External Pot: 1-10k ohms, 2 Watt minimum	

Control Output (Programmable Output, form C relay)
Resistive Rating: 3.0A at 30V DC, 125V AC and 240V AC Inductive Rating: 0.5A at 30V DC, 125V AC, and 240V AC

Recommended Fuses and Circuit Breakers
Fuse: UL Class J, CC, T or Type BS88; 600V (550V) or equivalent. Circuit Breakers: HMCP or Bulletin 140U or equivalent.

Protective Features
Motor Protection: 1 st overload protection - 150% for 60 Secs, 200% for 3 Secs (Provides Class 10 protection) Overcurrent: 200% hardware limit, 300% instantaneous fault

Over Voltage:	100-120V AC Input - Trip occurs at 405V DC bus voltage (equivalent to 150V AC incoming line) 200-240V AC Input - Trip occurs at 405V DC bus voltage (equivalent to 290V AC incoming line) 380-460V AC Input - Trip occurs at 810V DC bus voltage (equivalent to 575V AC incoming line)
Under Voltage:	100-120V AC Input - Trip occurs at 210V DC bus voltage (equivalent to 75V AC incoming line) 200-240V AC Input - Trip occurs at 210V DC bus voltage (equivalent to 150V AC incoming line) 380-480V AC Input - Trip occurs at 390V DC bus voltage (equivalent to 275V AC incoming line)

Control Ride Through: Minimum ride through is 0.5 Secs - typical value 2 Secs Faultless Power Ride Through: 100 milliseconds

Dynamic Braking
Internal brake IGBT included with all ratings except No Brake versions. Refer to Appendix B of the PowerFlex 4 User Manual on CD for ordering information.

(1) 200-240V AC - 1-Phase drives are also available with an integral EMC filter. Catalog suffix changes from N103 to N113 and N104 to N114.

(2) Refer to the Bulletin 140M Motor Protectors, Selection Guide, publication 140M-SG001... to determine the frame and breaking capacity required for your application.

Category	Specification	
Environment	Altitude:	1000 m (3300 ft) max. without derating
	Maximum Surrounding Air Temperature without derating:	
	IP20:	-10 to 50° C (14 to 122° F)
	NEMA 1:	-10 to 40° C (14 to 104° F)
	Flange Mount:	Heatsink: -10 to 40° C (14 to 104° F) Drive: -10 to 50° C (14 to 122° F)
	Cooling Method	
	Convection:	0.2 kW (0.25 HP) drives Flange Mount drives, all ratings
	Fan:	All other drive ratings.
	Storage Temperature:	-40 to 85 degrees C (-40 to 185 degrees F)
Control	Atmosphere:	Important: Drive must not be installed in an area where the ambient atmosphere contains volatile or corrosive gas, vapors or dust. If the drive is not going to be installed for a period of time, it must be stored in an area where it will not be exposed to a corrosive atmosphere.
	Relative Humidity:	0 to 95% non-condensing
	Shock (operating):	15G peak for 11ms duration (± 1.0 ms)
	Vibration (operating):	1G peak, 5 to 2000 Hz
	Carrier Frequency	2-16 kHz. Drive rating based on 4 kHz.
	Frequency Accuracy	
	Digital Input:	Within $\pm 0.05\%$ of set output frequency.
	Analog Input:	Within 0.5% of maximum output frequency.
	Speed Regulation - Open Loop with Slip Compensation:	$\pm 2\%$ of base speed across a 40:1 speed range.
	Stop Modes:	Multiple programmable stop modes including - Ramp, Coast, DC-Brake, Ramp-to-Hold and S Curve.
	Accel/Decel:	Two independently programmable accel and decel times. Each time may be programmed from 0 - 600 seconds in 0.1 second increments.
	Intermittent Overload:	150% Overload capability for up to 1 minute 200% Overload capability for up to 3 seconds
	Electronic Motor Overload Protection	Class 10 protection with speed sensitive response.

Notes:

Accessories and Dimensions

For information on...	See page
Product Selection	B-1
Product Dimensions	B-5

Product Selection

Table B.A Catalog Number Description

22A	-	A	1P5	N	1	1	4
Drive	Voltage Rating	Rating	Enclosure	HIM	Emission Class	Type	

Table B.B PowerFlex 4 Drives

Drive Ratings				Catalog Number		Frame Size
Input Voltage	kW	HP	Output Current	Panel Mount	Flange Mount ⁽¹⁾	
120V 50/60 Hz 1-Phase No Filter	0.2	0.25	1.5A	22A-V1P5N104	22A-V1P5F104	A
	0.4	0.5	2.3A	22A-V2P3N104	22A-V2P3F104	A
	0.75	1.0	4.5A	22A-V4P5N104	22A-V4P5F104	B
	1.1	1.5	6.0A	22A-V6P0N104	22A-V6P0F104	B
240V 50/60 Hz 1-Phase NO BRAKE With Integral "S Type" EMC Filter	0.2	0.25	1.4A	22A-A1P4N113	—	A
	0.4	0.5	2.1A	22A-A2P1N113	—	A
	0.75	1.0	3.6A	22A-A3P6N113	—	A
	1.5	2.0	6.8A	22A-A6P8N113	—	B
240V 50/60 Hz 1-Phase NO BRAKE No Filter	0.2	0.25	1.4A	22A-A1P4N103	22A-A1P4F103	A
	0.4	0.5	2.1A	22A-A2P1N103	22A-A2P1F103	A
	0.75	1.0	3.6A	22A-A3P6N103	22A-A3P6F103	A
	1.5	2.0	6.8A	22A-A6P8N103	22A-A6P8F103	B
240V 50/60 Hz 1-Phase With Integral "S Type" EMC Filter	0.2	0.25	1.5A	22A-A1P5N114	—	A
	0.4	0.5	2.3A	22A-A2P3N114	—	A
	0.75	1.0	4.5A	22A-A4P5N114	—	A
	1.5	2.0	8.0A	22A-A8P0N114	—	B
240V 50/60 Hz 1-Phase No Filter	0.2	0.25	1.5A	22A-A1P5N104	22A-A1P5F104	A
	0.4	0.5	2.3A	22A-A2P3N104	22A-A2P3F104	A
	0.75	1.0	4.5A	22A-A4P5N104	22A-A4P5F104	A
	1.5	2.0	8.0A	22A-A8P0N104	22A-A8P0F104	B
240V 50/60 Hz 3-Phase No Filter	0.2	0.25	1.5A	22A-B1P5N104	22A-B1P5F104	A
	0.4	0.5	2.3A	22A-B2P3N104	22A-B2P3F104	A
	0.75	1.0	4.5A	22A-B4P5N104	22A-B4P5F104	A
	1.5	2.0	8.0A	22A-B8P0N104	22A-B8P0F104	A
	2.2	3.0	12.0A	22A-B012N104	22A-B012F104	B
	3.7	5.0	17.5	22A-B017N104	22A-B017F104	B
480V 50/60 Hz 3-Phase No Filter	0.4	0.5	1.4A	22A-D1P4N104	22A-D1P4F104	A
	0.75	1.0	2.3A	22A-D2P3N104	22A-D2P3F104	A
	1.5	2.0	4.0A	22A-D4P0N104	22A-D4P0F104	A
	2.2	3.0	6.0A	22A-D6P0N104	22A-D6P0F104	B
	3.7	5.0	8.7A	22A-D8P7N104	22A-D8P7F104	B

⁽¹⁾ Replacement Plate drives are also available. Contact factory for ordering information.

Table B.C Dynamic Brake Modules

Drive Ratings				Catalog Number ⁽¹⁾
Input Voltage	kW	HP	Minimum Resistance Ω	
120V 50/60 Hz 1-Phase	0.2	0.25	60	AK-R2-091P500
	0.4	0.5	60	AK-R2-091P500
	0.75	1.0	60	AK-R2-091P500
	1.1	1.5	60	AK-R2-091P500
240V 50/60 Hz 1-Phase	0.2	0.25	60	AK-R2-091P500
	0.4	0.5	60	AK-R2-091P500
	0.75	1.0	60	AK-R2-091P500
	1.5	2.0	60	AK-R2-091P500
240V 50/60 Hz 3-Phase	0.2	0.25	60	AK-R2-091P500
	0.4	0.5	60	AK-R2-091P500
	0.75	1.0	60	AK-R2-091P500
	1.5	2.0	60	AK-R2-091P500
	2.2	3.0	48	AK-R2-047P500
	3.7	5.0	32	AK-R2-047P500
480V 50/60 Hz 3-Phase	0.4	0.5	121	AK-R2-091P500
	0.75	1.0	121	AK-R2-360P500
	1.5	2.0	121	AK-R2-360P500
	2.2	3.0	97	AK-R2-120P1K2
	3.7	5.0	97	AK-R2-120P1K2

⁽¹⁾ The resistors listed in this table are rated for 5% duty cycle.

Table B.D Bulletin 1321-3R Series Line Reactors

Input Voltage	kW	HP	Fundamental Amps	Maximum Continuous Amps	Inductance	Watts Loss	Catalog Number ⁽¹⁾
240V 50/60 Hz 3-Phase	0.2	0.25	2	3	12.0 mh	7.5 W	1321-3R2-A
	0.4	0.5	4	6	12.0 mh	21 W	1321-3R4-D
	0.75	1.0	8	12	3.0 mh	29 W	1321-3R8-B
	1.5	2.0	8	12	1.5 mh	19.5 W	1321-3R8-A
	2.2	3.0	12	18	1.25 mh	26 W	1321-3R12-A
	3.7	5.0	18	27	0.8 mh	36 W	1321-3R18-A
480V 50/60 Hz 3-Phase	0.4	0.5	2	3	20.0 mh	11.3 W	1321-3R2-B
	0.75	1.0	4	6	9.0 mh	20 W	1321-3R4-C
	1.5	2.0	4	6	6.5 mh	20 W	1321-3R4-B
	2.2	3.0	8	12	5.0 mh	25.3 W	1321-3R8-C
	3.7	5.0	18	27	3.0 mh	29 W	1321-3R-12-B

⁽¹⁾ Catalog numbers listed are for 3% impedance open style units. NEMA Type 1 and 5% impedance reactor types are also available. Refer to publication 1321-TD001....

Table B.E EMC Line Filters

Drive Ratings			S Type Filter Catalog Number ⁽¹⁾	L Type Filter Catalog Number ⁽³⁾
Input Voltage	kW	HP		
120V 50/60 Hz 1-Phase	0.2	0.25	–	22-RF010-AL
	0.4	0.5	–	22-RF010-AL
	0.75	1.0	–	22-RF018-BL
	1.1	1.5	–	22-RF025-CL ⁽⁴⁾
240V 50/60 Hz 1-Phase	0.2	0.25	⁽²⁾	22-RF010-AL
	0.4	0.5	⁽²⁾	22-RF010-AL
	0.75	1.0	⁽²⁾	22-RF010-AL
	1.5	2.0	⁽²⁾	22-RF018-BL
240V 50/60 Hz 1-Phase NO BRAKE	0.2	0.25	⁽²⁾	22-RF010-AL
	0.4	0.5	⁽²⁾	22-RF010-AL
	0.75	1.0	⁽²⁾	22-RF010-AL
	1.5	2.0	⁽²⁾	22-RF018-BL
240V 50/60 Hz 3-Phase	0.2	0.25	22-RF9P5-AS	22-RF9P5-AL
	0.4	0.5	22-RF9P5-AS	22-RF9P5-AL
	0.75	1.0	22-RF9P5-AS	22-RF9P5-AL
	1.5	2.0	22-RF9P5-AS	22-RF9P5-AL
	2.2	3.0	22-RF021-BS	22-RF021-BL
	3.7	5.0	22-RF021-BS	22-RF021-BL
480V 50/60 Hz 3-Phase	0.4	0.5	22-RF5P7-AS	22-RF5P7-AL
	0.75	1.0	22-RF5P7-AS	22-RF5P7-AL
	1.5	2.0	22-RF5P7-AS	22-RF5P7-AL
	2.2	3.0	22-RF012-BS	22-RF012-BL
	3.7	5.0	22-RF012-BS	22-RF012-BL

⁽¹⁾ This filter is suitable for use with a cable length up to 10 meters (33 feet) for Class A and 1 meter for Class B environments.

⁽²⁾ These ratings can be ordered with internal “S Type” filters. Refer to the Catalog Number explanation on [page P-4](#) and [Table B.B](#) for details.

⁽³⁾ This filter is suitable for use with a cable length up to 100 meters for Class A and 5 meters for Class B environments.

⁽⁴⁾ The piggyback mounting option cannot be used with Frame B PowerFlex 4 drives and Frame C EMC Line Filters.

Table B.F Human Interface Module (HIM) Option Kits and Accessories

Item	Description	Catalog Number
LCD Display, Remote Panel Mount	Digital speed control CopyCat capable IP66 (NEMA Type 4X/12) indoor use only 22-HIM-C2 includes 2.9 meter cable. 22-HIM-C2S includes 2 meter cable.	22-HIM-C2 22-HIM-C2S ⁽¹⁾
LCD Display, Remote Handheld	Digital speed control Full numeric keypad CopyCat capable IP30 (NEMA Type 1) Includes 1.0 meter cable Panel mount with optional Bezel Kit	22-HIM-A3
Bezel Kit	Panel mount for LCD Display, Remote Handheld unit, IP30 (NEMA Type 1)	22-HIM-B1
DSI HIM Cable (DSI HIM to RJ45 cable)	1.0 Meter (3.3 Feet) 2.9 Meter (9.51 Feet)	22-HIM-H10 22-HIM-H30

⁽¹⁾ The 22-HIM-C2S is smaller than the 22-HIM-C2 and cannot be used as a direct replacement.

Table B.G Communication Option Kits

Item	Description	Catalog Number ⁽²⁾
Serial Converter Module (RS485 to RS232)	Provides serial communication via DF1 protocol for use with DriveExplorer and DriveExecutive software. Includes: DSI to RS232 serial converter (1) 1203-SFC serial cable (1) 22-RJ45CBL-C20 cable (1) DriveExplorer Lite CD (1)	22-SCM-232
DSI Cable	2.0 meter RJ45 to RJ45 cable, male to male connectors.	22-RJ45CBL-C20
Serial Cable	2.0 meter serial cable with a locking low profile connector to connect to the serial converter and a 9-pin sub-miniature D female connector to connect to a computer.	1203-SFC
Null Cable Converter	For use when connecting the serial converter to DriveExplorer on a handheld PC.	1203-SNM
Splitter Cable	RJ45 one to two port splitter cable	AK-U0-RJ45-SC1
Terminating Resistors	RJ45 120 Ohm resistors (2 pieces)	AK-U0-RJ45-TR1
Terminal Block	RJ45 Two position terminal block (5 pieces)	AK-U0-RJ45-TB2P
DriveExplorer Software (CD-ROM) Version 3.01 or later	Windows based software package that provides an intuitive means for monitoring or configuring Allen-Bradley drives and communication adapters online. Compatibility: Windows 95, 98, ME, NT 4.0 (Service Pack 3 or later), 2000, XP and CE ⁽¹⁾	9306-4EXP01ENE
DriveExecutive software (CD-ROM) Version 1.01 or later	Windows based software package that provides an intuitive means for monitoring or configuring Allen-Bradley drives and communication adapters online and offline. Compatibility: Windows 98, ME, NT 4.0 (Service Pack 3 or later), 2000 and XP	9303-4DTE01ENE

⁽¹⁾ See www.ab.com/drives/driveexplorer.htm for supported devices.

⁽²⁾ For pricing information, refer to the PowerFlex 4-Class *Price List*, Publication 22-PL001....

Table B.H IP30/NEMA 1/UL Type 1 Kit

Item	Description	Drive Frame	Catalog Number ⁽¹⁾
IP30/NEMA 1/UL Type 1 Kit	Field installed kit. Converts drive to IP30/NEMA 1/UL Type 1 enclosure. Includes conduit box with mounting screws and plastic top panel.	A	22-JBAA
		B	22-JBAB

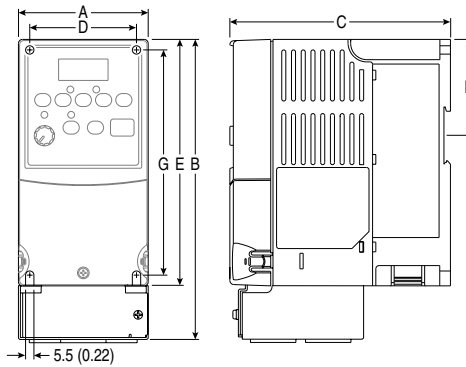
⁽¹⁾ For pricing information, refer to the PowerFlex 4-Class *Price List*, Publication 22-PL001....

Product Dimensions

Table B.1 PowerFlex 4 Panel Mount Drives – Ratings are in kW and (HP)

Frame	120V AC – 1-Phase	240V AC – 1-Phase No Brake	240V AC – 1-Phase	240V AC – 3-Phase	480V AC – 3-Phase
A	0.2 (0.25) 0.37 (0.5)	0.2 (0.25) 0.37 (0.5) 0.75 (1.0)	0.2 (0.25) 0.37 (0.5) 0.75 (1.0)	0.2 (0.25) 0.37 (0.5) 0.75 (1.0) 1.5 (2.0)	0.37 (0.5) 0.75 (1.0) 1.5 (2.0)
B	0.75 (1.0) 1.1 (1.5)	1.5 (2.0) 2.2 (3.0)	1.5 (2.0)	2.2 (3.0) 3.7 (5.0)	2.2 (3.0) 3.7 (5.0)

Figure B.1 PowerFlex 4 Panel Mount Drives – Dimensions are in millimeters and (inches). Weights are in kilograms and (pounds).



Frame	A	B ⁽¹⁾	C	D	E ⁽²⁾	F	G	Shipping Weight
A	80 (3.15)	185 (7.28)	136 (5.35)	67 (2.64)	152 (5.98)	59.3 (2.33)	140 (5.51)	1.4 (3.1)
B	100 (3.94)	213 (8.39)	136 (5.35)	87 (3.43)	180 (7.09)	87.4 (3.44)	168 (6.61)	2.2 (4.9)

⁽¹⁾ Overall height of drive with IP 30/NEMA 1/UL Type 1 option kit installed.

⁽²⁾ Overall height of standard IP 20/Open Type drive.

Figure B.2 IP 30/NEMA 1/UL Type 1 Option Kit – Dimensions are in millimeters and (inches)

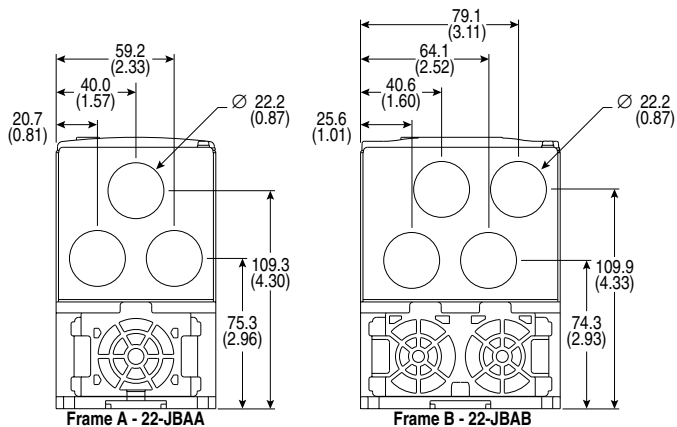


Figure B.3 PowerFlex 4 Flange Mount Drives – Dimensions are in millimeters and (inches)

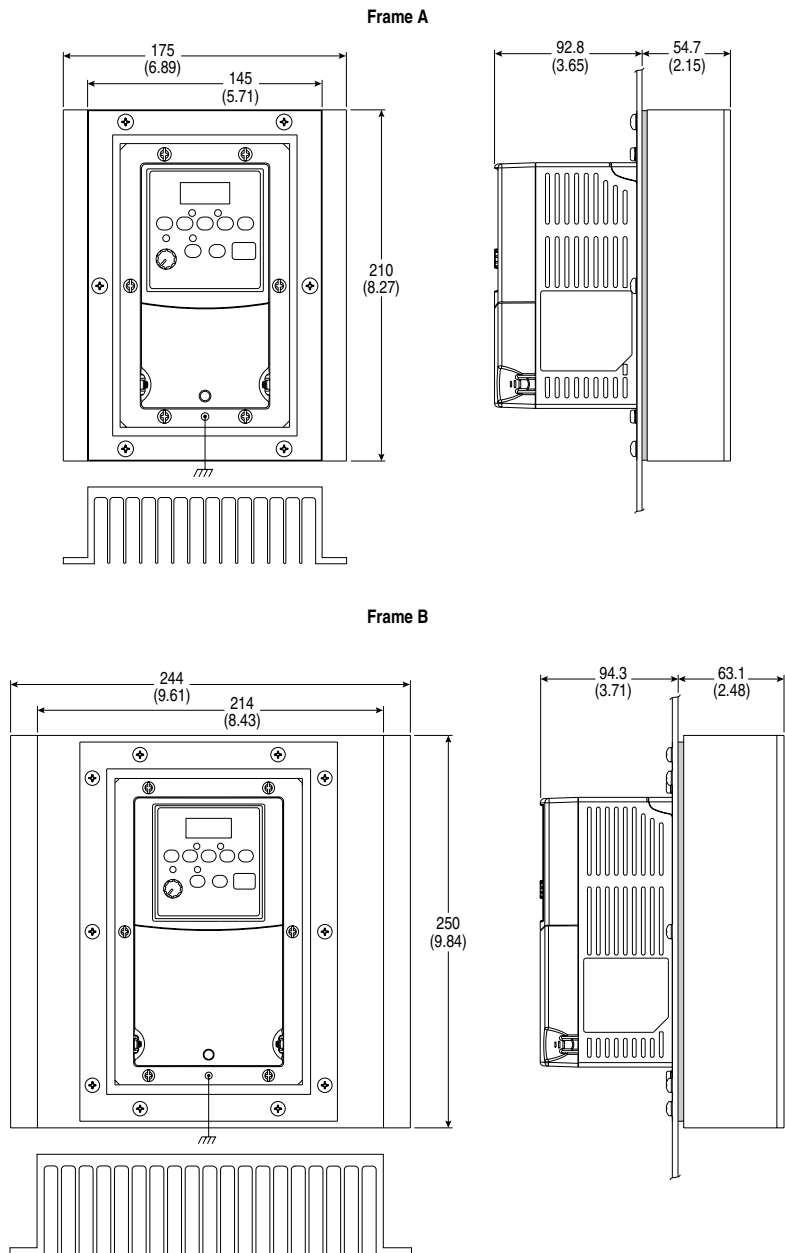


Figure B.4 PowerFlex 4 Cutout Dimensions – Dimensions are in millimeters and (inches)

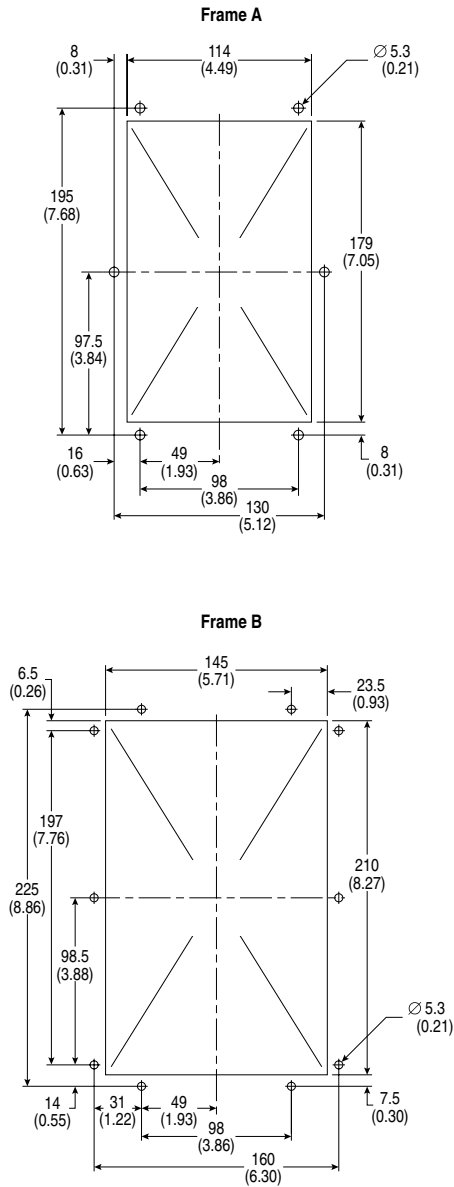
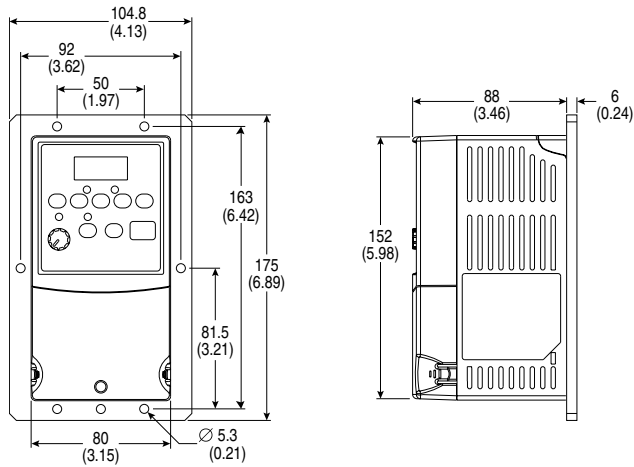


Figure B.5 PowerFlex 4 Replacement Plate Drive Dimensions – Dimensions are in millimeters and (inches)

Frame A



Frame B

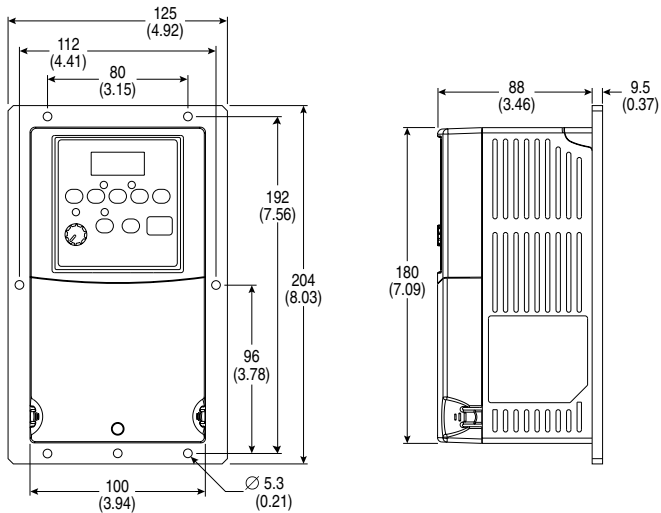
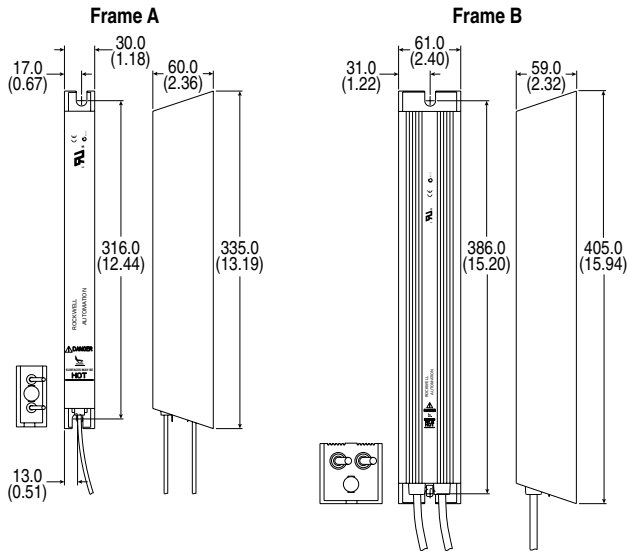
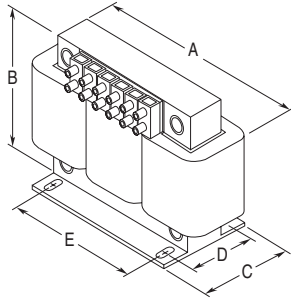


Figure B.6 Dynamic Brake Modules – Dimensions are in millimeters and (inches)

Frame	Catalog Number
A	AK-R2-091P500, AK-R2-047P500, AK-R2-360P500
B	AK-R2-030P1K2, AK-R2-120P1K2

Figure B.7 Bulletin 1321-3R Series Line Reactors

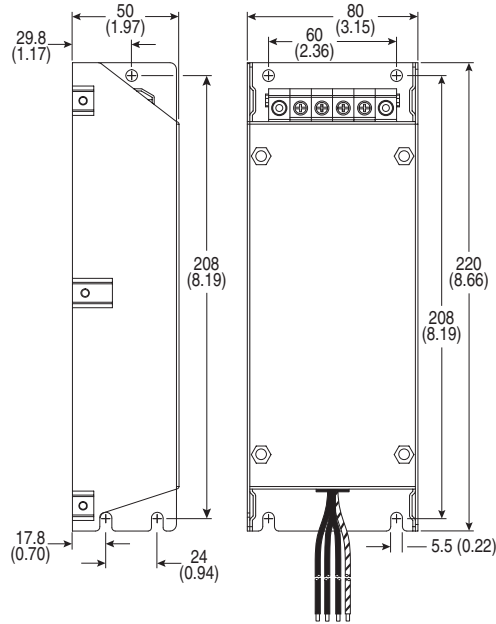
– Dimensions are in millimeters and (inches). Weights are in kilograms and (pounds).



Catalog Number	A	B	C	D	E	Weight
1321-3R2-A	112 (4.40)	104 (4.10)	70 (2.75)	50 (1.98)	37 (1.44)	1.8 (4)
1321-3R2-B	112 (4.40)	104 (4.10)	70 (2.75)	50 (1.98)	37 (1.44)	1.8 (4)
1321-3R4-B	112 (4.40)	104 (4.10)	76 (3.00)	50 (1.98)	37 (1.44)	1.8 (4)
1321-3R4-C	112 (4.40)	104 (4.10)	86 (3.38)	60 (2.35)	37 (1.44)	2.3 (5)
1321-3R4-D	112 (4.40)	104 (4.10)	92 (3.62)	66 (2.60)	37 (1.44)	2.7 (6)
1321-3R8-A	152 (6.00)	127 (5.00)	76 (3.00)	53 (2.10)	51 (2.00)	3.1 (7)
1321-3R8-B	152 (6.00)	127 (5.00)	76 (3.00)	53 (2.10)	51 (2.00)	3.6 (8)
1321-3R8-C	152 (6.00)	127 (5.00)	85 (3.35)	63 (2.48)	51 (2.00)	4.9 (11)
1321-3R12-A	152 (6.00)	127 (5.00)	76 (3.00)	53 (2.10)	51 (2.00)	4.1 (9)
1321-3R18-A	152 (6.00)	133 (5.25)	79 (3.10)	54 (2.13)	51 (2.00)	4.1 (9)

Figure B.8 Frame A EMC Line Filters – Dimensions are in millimeters and (inches)

Catalog Numbers: 22-RF5P7-AS, -AL; 22-RF9P5-AS, -AL; 22-RF010-AL

**Figure B.9 Frame B EMC Line Filters** – Dimensions are in millimeters and (inches)

Catalog Numbers: 22-RF012-BS, -BL; 22-RF018-BS; 22-RF021-BS, -BL

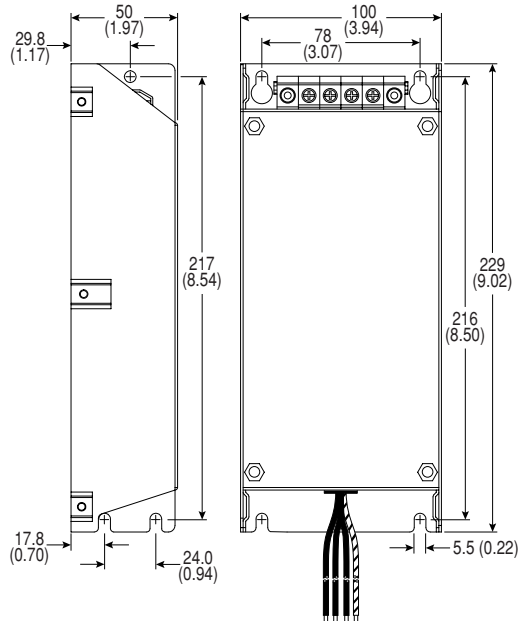
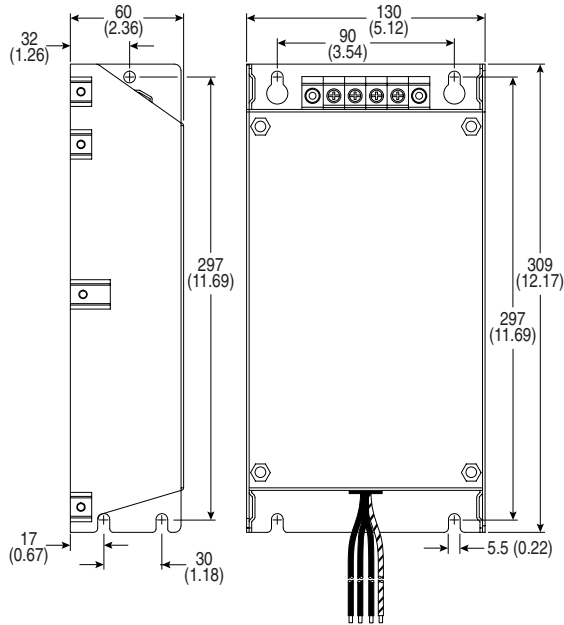


Figure B.10 Frame C EMC Line Filters – Dimensions are in millimeters and (inches)
Catalog Number: 22-RF025-CL



Important: The piggyback mounting option cannot be used with Frame B PowerFlex 4 drives and Frame C EMC Line Filters.

Figure B.11 Remote (Panel Mount) HIM – Dimensions are in millimeters and (inches)
Catalog Number: 22-HIM-C2

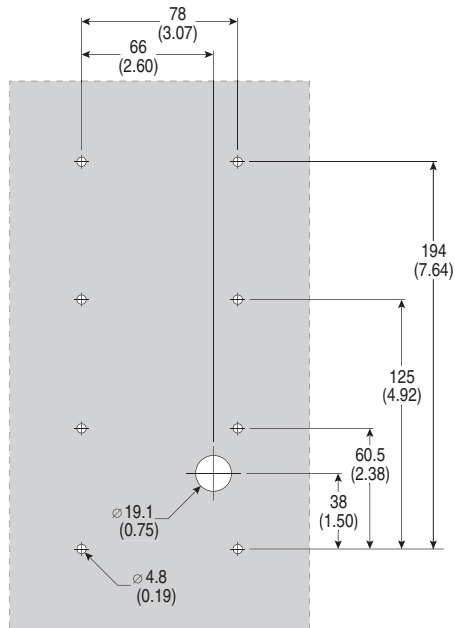
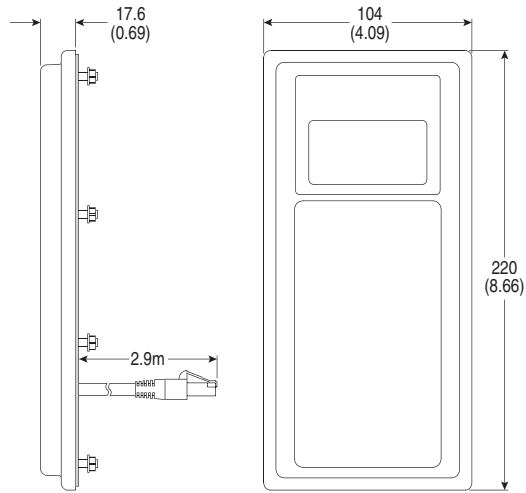
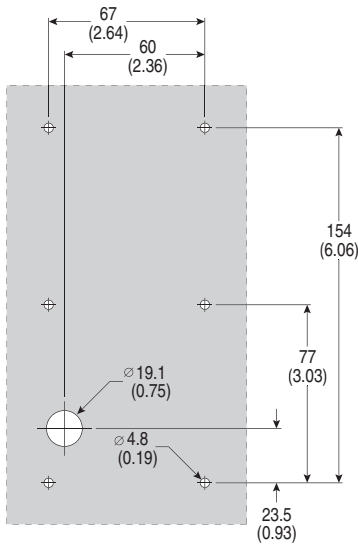
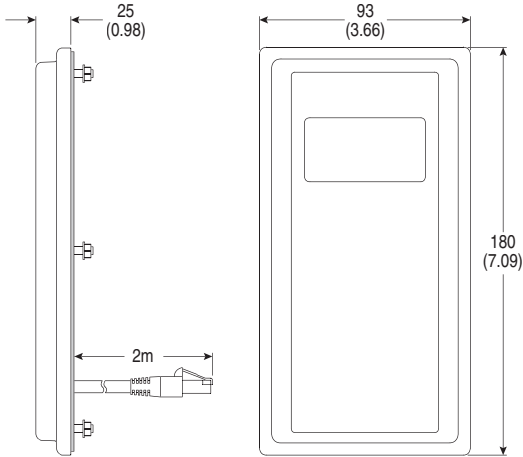
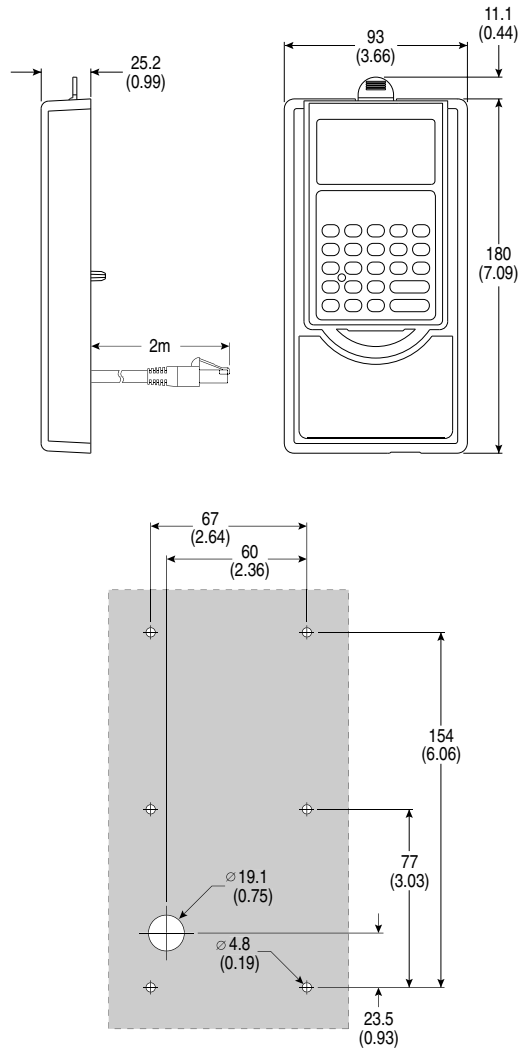


Figure B.12 Remote (Panel Mount) Small HIM – Dimensions are in millimeters and (inches) Catalog Number: 22-HIM-C2S



Important: The 22-HIM-C2S is smaller than the 22-HIM-C2 and cannot be used as a direct replacement.

Figure B.13 NEMA Type 1 Bezel – Dimensions are in millimeters and (inches)
Catalog Number: 22-HIM-B1



Notes: